Exam II
Review Topics

1. Derivative of $y = f(x)$ at a point $x = a$ 3.1:13-16,20-23(22,23)
   - a. Slope of the secant line to the graph of $y = f(x)$ over $[a, a+h]$
   - b. Slope of the tangent line to the graph of $y = f(x)$ at $x = a$
   - c. Average rate of change of $y = f(x)$ over $[a, a+h]$
   - d. Instantaneous rate of change of $y = f(x)$ at $x = a$
   - e. Difference quotient for $y = f(x)$ over $[a, a+h]$
   - f. Definition of derivative of $y = f(x)$ at $x = a$

2. Equation of the tangent line to $y = f(x)$ at a point $x = a$ 3.1:29-30; 3.2:27-31(28,30); 3.3:45-50(45,49); 3.5:41-45(41,45); 3.6:33-36(34)
   - a. Point-slope formula

3. Relationship between the graphs of $y = f(x)$ and $y' = f'(x)$ 3.1:6-9

4. (Non-)Existence of a derivative for a function $y = f(x)$ at a point $x = a$
   - a. Function f undefined at $x = a$
   - b. Function f discontinuous at $x = a$
   - c. Graph of $y = f(x)$ has a corner at $x = a$

5. Differentiability of $y = f(x)$ at $x = a$ implies continuity of $y = f(x)$ at $x = a$

6. Rules for differentiation 3.2:11-20(13,20)
   - a. Constant rule
   - b. Power rule
   - c. Sum rule
   - d. Difference rule
   - e. Product rule
   - f. Quotient rule

7. Derivatives
   - a. Polynomials
   - b. Rational Functions
   - c. Algebraic Functions

8. Higher Order Derivatives 3.2:21-26(26); 3.3:33-41(41);3.6:45-46(46)

9. Rules of Special Functions 3.3:9-26(10,17,26)

a. Rules for special functions

a. Sine function rule

b. Cosine function rule

c. Other trig function rules (via quotient rule)

d. Exponential function (e^x)

e. (Natural) logarithm function ($\ln x$)

10. Rectilinear Motion 3.4:17-20,36-37(18,36)

a.
$$\begin{array}{ll} \text{position function} & s = s(t) \\ \text{velocity function} & v = s'(t) \\ \text{acceleration funtion} & a = v'(t) = s''(t) \end{array}$$

b. Direction of motion determined by v

c. Total distance travelled

11. Projectile Motion 3.4:41-45(41,44)

a. $g \approx 32\ \text{ft}/\sec^2 \approx 9.8\ \text{m}/\sec^2$

b. $h(t) = s_0 + v_0 t - \frac{1}{2} g t^2$

c. $v = 0$ at maximum height

12. Chain rule 3.5:13-35(19,25,26,30)

a. Extended derivative rules

13. Implicit Differentiation 3.6:1-12(5,6,12)

a. General procedure

i. Equation $F(x, y) = 0$

ii. Treat (locally) y as a (unknown, but specified) function $f(x)$

iii. Differentiate the equation (both sides) w.r.t. x (writing y' for the derivative of y (the unknown, but specified function $f(x)$)

iv. Solve (linear) equation for y'

b. Tangent line construction

c. Higher order derivatives

14. Differentiation rules for special functions 3.6:19-25(20,21)

a. Roots ($y = \sqrt{x} = x^{1/2}$, $y = \sqrt[n]{x} = x^{1/n}$)

b. Powers ($y = x^r$)

c. Exponentials ($y = b^x$)

d. Logarithms ($y = \log_b x$)

e. Inverse trig functions

i. $y = \sin^{-1} x$

ii. $y = \tan^{-1} x$

iii. Others ($y = \sec^{-1} x$, $y = \cos^{-1} x$)

15. Logarithmic Differentiation 3.6:50-55(51,54)

a. Products, Quotients, Powers

b. Variable Base - Exponential Functions

16. Related Rate Problems 3.7:14-15,23-30(24,26)

a. Procedure (Page 158)